



QA/4HANA Studio
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PRODUCT DOCUMENTATION · CURRENT VERIFIED BUILD

Product Implementation, Backlog and Defect Status Report

A detailed evidence-based account of implemented functionality, every user story and acceptance criterion, remediated defects, remaining product gaps, and workspace-level capability.

DOCUMENT Q4H-PROD-STATUS-2.0	PRODUCT BASELINE QA/4HANA Studio 2.1.0 development line; v2.0.0 remains frozen
PREPARED 30 July 2026	CLASSIFICATION Internal product and training documentation

Screens containing configured target or account information have been anonymized. Passwords and credential values are never included.

DOCUMENT MAP

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Interpretation rule. "Implemented" means the tracker criterion is covered by repository evidence. "Partial" identifies delivered capability that still has acceptance work. "Open" is not yet accepted. "Deferred" is intentionally outside the current release scope.

CURRENT PRODUCT POSITION

1 Executive implementation summary

QA/4HANA Studio is a working, Canvas First SAP test-automation product spanning reusable authoring, controlled execution, immutable evidence and operational quality reporting.

41**Backlog items****29****Implemented****110/143****Accepted criteria****149****Catalogue tests****8****Partial items****2****Not started****2****Deferred****16/16****Defects remediated**

Delivered product spine

- Owner identity, encrypted SAP credential handling and authoritative target classification.
- Routeable Tests, datasets, Business Processes, Regression Packs, runs and audit records.
- Typed visual authoring from Single Test contracts through sequential hand-offs and independent Packs.
- Server-owned preflight, immutable effective-data snapshots, controlled execution and safe reruns.
- One canonical evidence PDF referenced by both Execution Center and Audit & Evidence.
- Accessible responsive shell, NVDA verification and release-governance evidence.

Current quality position

- Latest final source verification: 57 core, 44 isolated API and 40 isolated UI checks passed.
- Eight live-SAP checks remained deliberately gated in the final source verification.
- High-confidence secret scan passed across 259 non-ignored repository files.
- 20 recorded quality runs are retained in the current quality history.
- All 16 recorded historical failures have a remediation run; no open ledger defect remains.

Important distinction. The absence of an open recorded defect does not mean the backlog is complete. 12 items still contain partial, open or deferred acceptance scope and are listed in Section 5.

FUNCTIONAL INVENTORY

2 Detailed features by workspace

The following inventory describes current behavior rather than the original navigation labels. Business Processes and Regression Packs share one “Processes & Packs” workspace.

Automation Overview

A task-focused landing canvas for status, value, recent work and quick entry.

- Live counts for Tests, Business Processes, controls and evidence records.
- Recorded execution totals, pass/fail outcomes and automated runtime.
- Transparent modeled manual effort, potential time saved and cost-saved scenarios.
- Recent work cards and direct Create Test / New Execution actions.
- Authoritative configured-target state displayed in the shared header.

Remaining consideration: Exact recent-item deep links, readiness alerts, saved cost preferences and accessible historical trends remain.

Control Object Repository

Capture, curate and reuse SAP UI controls instead of duplicating selectors in Tests.

- Live-screen scan starts from the configured SAP target.
- Bulk scan and continuous Ctrl/Cmd interactive picking with duplicate detection.
- Domain/App ID browsing, local filtering, stable object routes and live highlight.
- Object edit, delete and keyboard-accessible ordering.
- Explicit App ID confirmation and curation before saving.

Remaining consideration: Persistent selector health/history, dependency-aware changes and contextual return to Compose remain.

Compose

Find, create, edit, validate and publish reusable Single Tests.

- Routeable Test Library with name, process-area, application and readiness filters.
- Guided blank-or-template creation with conflict-safe persistence.
- Visual typed Test contracts with input/output types and sensitivity.
- Step values distinguish literals, datasets, system context and prior outputs.
- Schema-driven ModuleCall editing, reusable object selection and keyboard reordering.
- Publish blocking for unresolved values, missing parameters/objects and output collisions.

Remaining consideration: No outstanding BL-020/BL-021 acceptance gap after the current reconciliation.

Test Data

Author reusable scalar, tabular, nested and relational transaction data.

- Flat CSV creation/editing with rich RFC4180 round-trip and dirty-state protection.
- Nested JSON transaction authoring for one or many headers with owned line items.
- Relational CSV builder with header/child keys and owned collection preview.
- Validation for duplicate keys, orphan children, missing keys and name collisions.
- Effective-data preview and immutable execution snapshots through preflight.

Remaining consideration: The full searchable library, typed/sensitive columns and dependency-safe rename/removal remain.

Processes & Packs

Model sequential business flows and independent regression collections visually.

- Business Processes sequence Tests and carry typed outputs into later stage inputs.
- Visual stage ordering, lifecycle, domain, application and data bindings.
- Validation for unresolved inputs, cycles, forward references, unknown/duplicate outputs and type mismatch.
- Regression Packs contain independent Test or Business Process members.
- Member-specific data, application, session and failure policies with draft/published lifecycle.
- Stable routes and direct execution of published mixed Saved Packs.

Remaining consideration: No outstanding BL-028/BL-029 acceptance gap after the current reconciliation.

Execution Center

Prepare, preflight, approve, run, monitor and safely rerun controlled automation.

- Execution types: Single Test, Business Process, Pack · Tests, Pack · Processes and Saved Pack.
- Four-stage flow: Scope, Data & policies, Preflight and Review.
- Server-owned preflight validates target, credentials, artifacts, versions, data, contracts and policies.
- Exact effective-data preview, mapping review, plan hash and immutable snapshot.
- Determinate progress for known work, persisted run routes and reconnect recovery.
- Fail-stop transaction behavior retains earlier evidence; no automatic reversal.
- Rerun difference review, immutable lineage and failed-scope safety restrictions.

Remaining consideration: Complete six-level event persistence/reconciliation and exact terminal-failure correction links remain.

Audit & Evidence

Search immutable execution history and open one canonical evidence record per run.

- Flat newest-first run library rather than date-folder navigation.
- Search and client filters for outcome, mode and date.
- Stable audit detail routes with executor, target and source context.
- Canonical evidence PDF shared with Execution Center—no duplicate evidence document.
- Evidence branding, Test name headers, redaction and retention governance.
- Preserved rerun lineage and immutable execution identifiers.

Remaining consideration: Server-side multi-field filtering, paging/sorting and full parent/child lineage traversal remain.

Settings, identity and account menu

Control owner identity, appearance, help and test-system connections.

- Single preserved workspace-owner identity with Google registration/sign-in support.
- Profile menu provides Settings, Help and Sign out.
- Appearance selection is managed inside Settings.
- SAP connection captures target URL and credentials without returning passwords to the browser.
- OS credential vault where available and encrypted local fallback for detached Windows sessions.
- Salesforce, Oracle and ServiceNow connection panels retained as future configuration surfaces.

Remaining consideration: Multi-user roles remain intentionally deferred; only SAP is integrated with execution and live scan.

QA/4HANA Test Operations companion app

Provide release-quality visibility across the automated product test estate.

- Repository-generated 149-test catalogue grouped by feature and workspace.
- Latest status, execution mode and recorded duration per test.
- Historical execution dashboard backed by recorded runs.
- Persistent failure ledger including failures later remediated and passed.

Remaining consideration: It is an operational companion rather than a primary Studio workspace.

41-ITEM SOURCE OF TRUTH

3

Granular product backlog and acceptance status

Every consolidated backlog item is shown with its description, user story, release/priority metadata, criterion-by-criterion status, implementation coverage and evidence.

Status	Items	Meaning
Implemented	29	All recorded acceptance criteria are satisfied.
Partial	8	Some functionality exists; one or more criteria still need delivery.
Not started	2	No acceptance criterion is complete.
Deferred	2	Deliberately outside the present release scope.

BL-001 · PB-001 · IMM-01

Implemented

Show verified execution context

PRIORITY P0	PLANNED TRANCHE T0 · Delivered baseline	ACTUAL RELEASE Not yet delivered	WORKSPACE Shell, Execution Center
ACCEPTANCE 3/3			

Description

Display only server-backed workspace and target context and avoid unsupported connection claims.

User story

As a tester, I want verified target context so that I know where an execution will act.

Acceptance criteria

- 1

Shell and review use server-backed target metadata.

Implemented
- 2

Unavailable or stale context is explicit and blocks unsafe execution.

Implemented
- 3

No decorative control implies a successful connection.

Implemented

Implementation coverage

Unsupported environment claims were removed; server-backed SAP configuration is displayed. Freshness verification remains strengthened by BL-014.

Evidence: App.tsx target label · P0_IMPLEMENTATION_REPORT.md
Last reconciled: 2026-07-27

BL-002 · PB-002 · IMM-02

Implemented

Review execution impact before Start

PRIORITY P0	PLANNED TRANCHE T0 · Delivered baseline	ACTUAL RELEASE Not yet delivered	WORKSPACE Execution Center
ACCEPTANCE 3/3			

Description

Present target, scope, data, side effects and evidence behavior before starting execution.

User story

As an operator, I want an impact review so that I can prevent unintended SAP changes.

Acceptance criteria

1	Review shows target, mode, ordered scope, data and browser policy.	Implemented
2	SAP side effects and cancellation boundary are explicit.	Implemented
3	Duplicate submission produces one run ID.	Implemented

Implementation coverage

Review, warning acknowledgement and duplicate-submit protection are implemented; authoritative matrix enhancements continue in BL-030.

Evidence: RunPanel.tsx · regression/ui/run-tab.test.js
Last reconciled: 2026-07-27

BL-003 · PB-003 · IMM-03

Implemented

Use truthful dashboard states

PRIORITY P0	PLANNED TRANCHE T0 · Delivered baseline	ACTUAL RELEASE Not yet delivered	WORKSPACE Automation Overview
ACCEPTANCE 3/3			

Description

Source Overview values from APIs and distinguish loading, empty and unavailable data.

User story

As a test manager, I want trustworthy dashboard states so that I do not act on fabricated information.

Acceptance criteria

1	Metrics and recent records come from APIs.	Implemented
2	Loading, empty and unavailable states are distinct.	Implemented
3	No fallback execution or document record is fabricated.	Implemented

Implementation coverage

Workspace counts, immutable runs and evidence are API-backed with retry and empty states.

Evidence: AutomationOverview.tsx · regression/ui/overview.test.js

Last reconciled: 2026-07-27

BL-004 · PB-004 · IMM-09

Implemented

Restrict local Studio exposure

PRIORITY P0	PLANNED TRANCHE T0 · Delivered baseline	ACTUAL RELEASE Not yet delivered	WORKSPACE Platform
ACCEPTANCE 3/3			

Description

Bind locally by default and require an explicit decision before network exposure.

User story

As the workspace owner, I want Studio locally restricted so that unfinished controls are not exposed.

Acceptance criteria

1	Default host is 127.0.0.1.	Implemented
2	Non-loopback binding requires an explicit override.	Implemented
3	Startup communicates network exposure risk.	Implemented

Implementation coverage

Loopback is the default and host override is explicit.

Evidence: packages/studio-server/src/standalone.ts · P0_IMPLEMENTATION_REPORT.md

Last reconciled: 2026-07-27

BL-005 · PB-005 · IMM-04

Implemented

Restore functional workspace styling

PRIORITY P0	PLANNED TRANCHE T0 · Delivered baseline	ACTUAL RELEASE Not yet delivered	WORKSPACE All workspaces
ACCEPTANCE 3/3			

Description

Provide coherent forms, tables, statuses, dialogs, drawers and responsive functional layouts.

User story

As a user, I want consistent functional styling so that every workspace is readable and operable.

Acceptance criteria

1	Functional classes have visible states and spacing.	Implemented
2	Status and feedback are not color-only.	Implemented
3	Core screens remain usable at supported widths.	Implemented

Implementation coverage

Canvas First shell and functional compatibility styles are present; component-system consolidation remains BL-011.

Evidence: packages/studio-web/src/index.css · P0_IMPLEMENTATION_REPORT.md

Last reconciled: 2026-07-27

BL-006 · PB-006 · IMM-05

Implemented

Preserve correct UI encoding

PRIORITY P0	PLANNED TRANCHE T0 · Delivered baseline	ACTUAL RELEASE Not yet delivered	WORKSPACE All workspaces
ACCEPTANCE 3/3			

Description

Keep source content UTF-8 and prevent accidental character corruption.

User story

As a user, I want readable product copy so that technical symbols and punctuation are unambiguous.

Acceptance criteria

1	Source is valid UTF-8.	Implemented
2	Build and browser output preserve characters.	Implemented
3	Display-tool mojibake is not written back to source.	Implemented

Implementation coverage

Source verification closed the issue; PowerShell display encoding was identified as the false signal.

Evidence: PRODUCT_BACKLOG.md P0 status

Last reconciled: 2026-07-27

BL-007 · PB-007 · IMM-07

Implemented

Protect unsaved artifact work

PRIORITY P0	PLANNED TRANCHE T0 · Delivered baseline	ACTUAL RELEASE Not yet delivered	WORKSPACE Compose, Test Data, Processes & Packs, Execution Center
ACCEPTANCE 3/3			

Description

Guard dirty tests, datasets, processes and execution drafts during navigation and reload.

User story

As an author, I want unsaved changes protected so that navigation does not destroy my work.

Acceptance criteria

1	Dirty state reaches shell navigation.	Implemented
2	Artifact switching and browser unload require confirmation.	Implemented
3	New drafts are dirty from their first meaningful change.	Implemented

Implementation coverage

Composer, Dataset, Group and Execution draft guards are implemented and regression tested.

Evidence: App.tsx navigation guard · regression/ui/unsaved-guards.test.js

Last reconciled: 2026-07-27

BL-008 · PB-008 · IMM-06

Implemented

Provide baseline keyboard and screen-reader support

PRIORITY P0	PLANNED TRANCHE T0 · Delivered baseline	ACTUAL RELEASE Not yet delivered	WORKSPACE All workspaces
ACCEPTANCE 3/3			

Description

Make core shell navigation, forms, actions and feedback semantically operable.

User story

As a keyboard or screen-reader user, I want equivalent access to core Studio actions.

Acceptance criteria

1	Primary navigation and actions use semantic controls.	Implemented
2	Focus is visible and labels are programmatic.	Implemented
3	Errors and status changes are announced where required.	Implemented

Implementation coverage

P0 semantic baseline is present; complex-widget behavior and full accessibility/release verification were subsequently closed by BL-016 and BL-017.

Evidence: App.tsx skip link and headings · P0_IMPLEMENTATION_REPORT.md
Last reconciled: 2026-07-27

BL-009 · PB-009 · IMM-10

Implemented

Run isolated frontend regression tests

PRIORITY P0	PLANNED TRANCHE T0 · Delivered baseline	ACTUAL RELEASE Not yet delivered	WORKSPACE Quality Engineering
ACCEPTANCE 3/3			

Description

Exercise UI workflows against temporary synthetic fixtures without live SAP side effects.

User story

As a developer, I want isolated UI regression coverage so that changes can be verified safely.

Acceptance criteria

1	Harness uses temporary storage.	Implemented
2	Execution is disabled unless explicitly approved.	Implemented
3	Fixtures and temporary databases are removed after the run.	Implemented

Implementation coverage

Isolated UI harness covers Overview, Compose, Data, Groups, Execution, Audit and dirty guards.

Evidence: regression/run-isolated-ui.js · regression/ui

Last reconciled: 2026-07-27

BL-010 · PB-010 · IMM-08

Implemented

Recover live monitoring connections

PRIORITY P0	PLANNED TRANCHE T0 · Delivered baseline	ACTUAL RELEASE Not yet delivered	WORKSPACE Execution Center
ACCEPTANCE 3/3			

Description

Reconnect polling safely and preserve run identity after temporary server failures.

User story

As an operator, I want monitoring to recover so that a network interruption does not hide a live run.

Acceptance criteria

1	Polling has bounded backoff and cancellation.	Implemented
2	Last update and connection state are visible.	Implemented
3	Refresh never starts duplicate execution work.	Implemented

Implementation coverage

Recursive polling, backoff, manual retry and cleanup are implemented.

Evidence: RunPanel.tsx · P0_IMPLEMENTATION_REPORT.md

Last reconciled: 2026-07-27

BL-011 · PB-011 · FND-01 · FND-02

Implemented

Consolidate design tokens and workspace primitives

PRIORITY P1	PLANNED TRANCHE T1 · 2.0 Alpha	ACTUAL RELEASE Not yet delivered	WORKSPACE Design System, All workspaces
ACCEPTANCE 3/3			

Description

Replace local styling variations with reusable Canvas First layout and interaction primitives.

User story

As a product team, we want shared primitives so that workspaces remain visually and behaviorally consistent.

Acceptance criteria

- 1

Tokens cover color, type, spacing, focus, status and elevation.

Implemented
- 2

Page header, toolbar, card, table, drawer and empty-state primitives are reused.

Implemented
- 3

Light and dark themes meet contrast requirements.

Implemented

Implementation coverage

Canvas First now exposes shared type, spacing, focus, status, elevation and theme tokens plus reusable page header, toolbar, card, table, drawer, empty and async-feedback primitives. Automated checks cover normal-text contrast in both themes.

Evidence: index.css T1.3 Canvas First primitives · WorkspacePrimitives.tsx · DocumentsPanel.tsx primitive adoption · t1-foundations.test.js theme contrast

Last reconciled: 2026-07-29

BL-012 · PB-012 · FND-03

Implemented

Provide stable artifact and run routes

PRIORITY P1	PLANNED TRANCHE T1 · 2.0 Alpha	ACTUAL RELEASE 2.1.0	WORKSPACE Shell, All workspaces
ACCEPTANCE 3/3			

Description

Give every workspace, artifact and run a recoverable URL with navigation guards.

User story

As a user, I want stable URLs so that refresh, bookmarks and correction links preserve context.

Acceptance criteria

- 1

Workspace routes survive refresh and browser navigation.

Implemented
- 2

Tests, datasets, processes, packs and audit records have stable detail routes.

Implemented
- 3

Dirty-state guards apply to route changes.

Implemented

Implementation coverage

Canonical routes restore Tests, datasets, Business Processes, Regression Packs, object selections, execution runs and audit records across refresh and browser back/forward. Preflight findings open the exact dataset/source or SAP setting, and shell, tab, browser-history and unload navigation protect unsaved work.

Evidence: routes.ts · App.tsx route orchestration · ProcessPacksWorkspace.tsx · PackEditor.tsx · DocumentsPanel audit detail · executionPreflight.ts
correctionRoute · stable-routes.test.js · packs-tab.test.js · unsaved-guards.test.js
Last reconciled: 2026-07-30

BL-013 · PB-013 · FND-04

Implemented

Complete the responsive application shell

PRIORITY P1	PLANNED TRANCHE T1 · 2.0 Alpha	ACTUAL RELEASE Not yet delivered	WORKSPACE Shell
ACCEPTANCE 3/3			

Description

Keep stable navigation, account actions and workspace context usable across supported widths.

User story

As a user, I want a calm responsive shell so that navigation does not dominate the work canvas.

Acceptance criteria

- 1

Sidebar collapses without hiding destinations.

Implemented
- 2

Account menu owns Settings, Help and Sign out.

Implemented
- 3

Header actions wrap or simplify at narrow widths without overlap.

Implemented

Implementation coverage

The compact sidebar retains programmatic destination names, the account menu owns Settings, Help and Sign out with complete menu-key behavior, and target plus workflow actions reflow without shell overflow at narrow widths.

Evidence: App.tsx responsive shell · AccountMenu.tsx menu keyboard contract · index.css narrow shell rules · t1-foundations.test.js narrow-width journey
Last reconciled: 2026-07-29

BL-014 · PB-014 · CTX-01

Implemented

Provide authoritative workspace and environment context

PRIORITY P0	PLANNED TRANCHE T1 · 2.0 Alpha	ACTUAL RELEASE Not yet delivered	WORKSPACE Shell, Settings, Control Object Repository, Execution Center
ACCEPTANCE 3/3			

Description

Unify owner, SAP target, safety class, profile reference and verification freshness.

User story

As an operator, I want one authoritative context so that every workspace refers to the same target.

Acceptance criteria

1	Server returns non-secret workspace and target context.	Implemented
2	Shell, scan and execution review show the same context and freshness.	Implemented
3	A stale or mismatched context blocks Start with recovery guidance.	Implemented

Implementation coverage

One non-secret server context now supplies owner, SAP hostname, profile, safety class and 24-hour verification freshness to the shell, live scanner and execution review. Preflight hashes that context, blocks unknown/stale targets and invalidates mismatches before Start.

Evidence: executionContext.ts · workspaceGovernance.ts · /api/workspace-context · ObjectScanner.tsx · RunPanel.tsx · execution-preflight.test.js · t1-security-context.test.js
Last reconciled: 2026-07-29

BL-015 · PB-015 · FND-05

Implemented

Standardise async and feedback states

PRIORITY P1	PLANNED TRANCHE T1 · 2.0 Alpha	ACTUAL RELEASE Not yet delivered	WORKSPACE All workspaces
ACCEPTANCE 3/3			

Description

Use consistent loading, empty, warning, error, success and retry behavior.

User story

As a user, I want predictable feedback so that I know whether Studio is working, waiting or blocked.

Acceptance criteria

- 1

Every API-backed surface has loading, empty and error states.

Implemented
- 2

Errors are actionable and safe.

Implemented
- 3

Success is associated with the item or action that changed.

Implemented

Implementation coverage

API-backed workspaces now use consistent announced loading, empty, actionable error/retry and item-associated save-success behavior. Existing execution, scan and settings feedback uses the same status semantics.

Evidence: WorkspacePrimitives.tsx AsyncFeedback and EmptyState · TestCaseEditor.tsx · DataEditor.tsx · GroupEditor.tsx · DocumentsPanel.tsx · ObjectBrowser.tsx
Last reconciled: 2026-07-29

BL-016 · PB-016 · FND-06 · FND-07

Implemented

Upgrade complex pickers, ordering and tables

PRIORITY P1	PLANNED TRANCHE T1 · 2.0 Alpha	ACTUAL RELEASE Not yet delivered	WORKSPACE Compose, Test Data, Processes & Packs, Control Object Repository, Execution Center
ACCEPTANCE 3/3			

Description

Implement accessible search, grouped selection, ordered transfer, drag alternatives and responsive tables.

User story

As a keyboard or assistive-technology user, I want complete complex-widget behavior.

Acceptance criteria

- 1

Grouped pickers implement an approved combobox or tree pattern.

Implemented
- 2

Every reorder action has keyboard controls and announcements.

Implemented
- 3

Tables preserve labels and actions at supported narrow widths.

Implemented

Implementation coverage

Grouped selection now uses a searchable combobox/listbox with Arrow, Home, End, Enter and Escape behavior. All draggable ordering surfaces also provide keyboard buttons and live announcements. Responsive tables expose row labels and retain actions.

Evidence: GroupedPicker.tsx · FileChainPicker.tsx · TestCaseEditor.tsx reorder alternatives · ObjectBrowser.tsx reorder alternatives · index.css responsive-table · t1-foundations.test.js keyboard journeys

Last reconciled: 2026-07-29

BL-017 · PB-031 · FND-09 · A11Y-01 · EXC-026 · EXC2-006

Implemented

Pass accessibility and production release verification

PRIORITY P0	PLANNED TRANCHE T4 · 2.0 GA	ACTUAL RELEASE Not yet delivered	WORKSPACE Quality Engineering, All workspaces
ACCEPTANCE 3/3			

Description

Validate completed journeys to WCAG 2.2 AA and assemble production release evidence.

User story

As a product owner, I want objective accessibility and release evidence before approving 2.0.

Acceptance criteria

1	Axe has no unapproved serious or critical findings.	Implemented
2	Keyboard and NVDA journeys are recorded for every primary workspace.	Implemented
3	Responsive, security, secret-scan and approved live SAP evidence are signed.	Implemented

Implementation coverage

All seven primary workspaces pass Axe with no serious/critical findings, keyboard skip/focus entry, and 320 CSS-pixel reflow. A paced NVDA 2026.1.1 session in headed Chrome 150 recorded every workspace landmark, heading, control role, label and state against isolated data. A Chrome-only 6px Execution Center overflow was found, fixed and retested. The final GA gate passed build, typecheck, 57 core tests, 28 isolated API tests, 32 isolated UI tests and a 256-file pre-cleanup secret scan; the frozen 251-file release set was rescanned successfully. Technical accessibility, security and positive P2P/O2C evidence verification was signed by the verification executor and accepted by workspace owner Sathyanarayanan Kiran at 21:18:11 +05:30. BL-041's technical hold was resolved, and the workspace owner approved QA/4HANA Studio 2.0 for GA on 30 July 2026 at 08:00:18 +05:30.

Evidence: regression/ui/accessibility.test.js · regression/results/nvda/primary-workspaces-2026-07-29.log · NVDA_PRIMARY_WORKSPACE_RESULTS.md · ACCESSIBILITY_RELEASE_VERIFICATION.md · RELEASE_SIGN_OFF_2.0.md
Last reconciled: 2026-07-30

BL-018 · PB-017 · DASH-01

Partial

Complete the task-focused Automation Overview

PRIORITY P1	PLANNED TRANCHE T4 · 2.0 GA	ACTUAL RELEASE Not yet delivered	WORKSPACE Automation Overview
ACCEPTANCE 1/3			

Description

Turn Overview into a trusted landing page with direct continuation, alerts and scoped metrics.

User story

As a test manager, I want Overview to show recent work and issues needing attention.

Acceptance criteria

1	Selected tests and runs open their exact detail routes.	Open
2	Configured target status is accurate and actionable.	Implemented
3	Recent failures, stale drafts and blocked readiness are surfaced without duplicating navigation.	Open

Implementation coverage

Canvas, API counts, recent work and impact dashboard exist, and target state now comes from authoritative workspace context. Recent rows still open the general workspace rather than exact details, and failure/readiness/draft alerts remain.

Evidence: AutomationOverview.tsx

Last reconciled: 2026-07-29

BL-019 · PB-034 · FUT-03

Partial

Add filterable execution value and trend analytics

PRIORITY
P1

PLANNED TRANCHE
T4 · 2.0 GA

ACTUAL RELEASE
Not yet delivered

WORKSPACE
Automation Overview

ACCEPTANCE
0/3

Description

Scope outcomes, runtime, manual effort and TCO by time, target, process and test.

User story

As a test manager, I want transparent trends so that I can identify risk and realised automation value.

Acceptance criteria

- 1

Every metric states date range, filters, numerator, denominator and freshness.

Open
- 2

Cost assumptions are saved as owner workspace preferences.

Open
- 3

Charts have accessible tabular equivalents and never fabricate trends.

Open

Implementation coverage

Actual totals and a broad TCO model exist, but metric scope/freshness, saved assumptions, filters, time-series trends and accessible chart equivalents are not complete.

Evidence: AutomationOverview.tsx ExecutionImpactDashboard
Last reconciled: 2026-07-29

BL-020 · PB-018 · TEST-01

Implemented

Provide a routeable Test Library

PRIORITY P1	PLANNED TRANCHE T2 · 2.0 Beta	ACTUAL RELEASE 2.1.0	WORKSPACE Compose
ACCEPTANCE 3/3			

Description

Find, create, duplicate, archive and open tests using business metadata rather than file-first controls.

User story

As a test author, I want a searchable Test Library so that I avoid duplicates and resume exact work.

Acceptance criteria

1	List supports name, process area, application and status filters.	Implemented
2	New Test captures business name, application, domain and blank/template choice.	Implemented
3	Opening or creating uses a stable test route while preserving JSON compatibility.	Implemented

Implementation coverage

Compose now opens a responsive Test Library with name/file search plus process-area, application and readiness filters. Guided creation captures the business name, generated or edited file name, application, process area and blank-versus-existing-Test starting point. Creation is server-persisted, conflict-safe and immediately opens a refresh-stable detail route; legacy Tests without application metadata remain valid and default to SAP in the library index.

Evidence: TestLibrary.tsx · TestCaseEditor.tsx application metadata · server.ts /api/testcases/library and POST /api/testcases/:file · api.ts listTestLibrary/createTestCase · testcases.test.js · test-library.test.js · compose.test.js · stable-routes.test.js · unsaved-guards.test.js

Last reconciled: 2026-07-30

BL-021 · PB-019 · TEST-02 · EXC-006 · EXC-007 · EXC2-003 · SC-01 · SC-02 · SC-03

Implemented

Build typed, validated Single Tests

PRIORITY
P1

PLANNED TRANCHE
T2 · 2.0 Beta

ACTUAL RELEASE
2.1.0

WORKSPACE
Compose

ACCEPTANCE
4/4

Description

Compose steps, objects, value sources, declared inputs/outputs and readiness in one guided canvas.

User story

As a functional consultant, I want guided test composition so that saved Tests are understandable and runnable.

Acceptance criteria

- 1

Steps distinguish literal, dataset, system-context and prior-output values.

Implemented
- 2

Tests visually declare typed inputs, outputs and sensitivity.

Implemented
- 3

Required objects, parameters, unresolved values and output collisions block publishing.

Implemented
- 4

Reordering is keyboard accessible and existing ModuleCall JSON round-trips.

Implemented

Implementation coverage

Compose now distinguishes literal, dataset, system-context and prior-output step values; visually authors typed Test inputs, outputs and sensitivity; and blocks publishing for unresolved required values, missing objects or parameters, and output collisions. Existing ModuleCall JSON remains executable and round-trips through the editor, while keyboard-accessible reordering is retained.

Evidence: TestCaseEditor.tsx · StepEditor.tsx · TestContractEditor.tsx · packages/core/src/domain/testContract.ts · server.ts Test publish validation · testcases.test.js · typed-test-authoring.test.js · compose.test.js

Last reconciled: 2026-07-30

BL-022 · PB-020 · OBJ-01

Partial

Complete the Control Object Repository workbench

PRIORITY
P1

PLANNED TRANCHE
T2 · 2.0 Beta

ACTUAL RELEASE
Not yet delivered

WORKSPACE
Control Object Repository

ACCEPTANCE
1/4

Description

Search, inspect, edit, highlight and safely maintain reusable SAP UI5 objects.

User story

As an automation engineer, I want a central object workbench so that selectors remain reusable and understandable.

Acceptance criteria

- 1

Search spans domain, App ID, name, label, type and stability.

Partial
- 2

Object detail shows selector metadata, scope, usage and last verification.

Partial
- 3

Rename and delete show dependency impact and preserve or guide reference correction.

Open
- 4

Ordering has keyboard controls and does not require drag.

Implemented

Implementation coverage

Domain/App ID browse, local filtering, stable object routes, highlight, edit/delete and keyboard ordering exist. Stability, usage/verification metadata and dependency-safe rename/delete remain.

Evidence: ObjectBrowser.tsx · server object APIs
Last reconciled: 2026-07-29

BL-023 · PB-021 · OBJ-01

Partial

Complete guided live-screen capture

PRIORITY P1	PLANNED TRANCHE T2 · 2.0 Beta	ACTUAL RELEASE Not yet delivered	WORKSPACE Control Object Repository, Compose
ACCEPTANCE 3/4			

Description

Verify the SAP target, capture or pick controls, curate names and save stable objects safely.

User story

As a functional consultant, I want guided capture so that objects are stable and stored under the correct application.

Acceptance criteria

1	Scan starts from the configured SAP URL and displays target/session health.	Implemented
2	Bulk capture and interactive picking identify exact duplicates before save.	Implemented
3	App ID is detected or explicitly confirmed for each capture.	Implemented
4	Contextual capture returns the saved object to the originating Compose field without losing work.	Open

Implementation coverage

Configured SAP URL/context health, session recovery, bulk capture, continuous Ctrl/Cmd picking, exact duplicate detection, explicit App ID, highlighting and Save all exist. Contextual return to the originating Compose field remains.

Evidence: ObjectScanner.tsx · CurationList.tsx · scanSession.ts
Last reconciled: 2026-07-29

BL-024 · New · Control Object Repository stocktake

Not started

Add selector health, history and governance

PRIORITY
P1

PLANNED TRANCHE
T2 · 2.0 Beta

ACTUAL RELEASE
Not yet delivered

WORKSPACE
Control Object Repository, Audit & Evidence

ACCEPTANCE
0/4

Description

Track selector quality, verification history, duplicates and change impact over time.

User story

As an automation maintainer, I want selector health and lineage so that object changes do not silently break Tests.

Acceptance criteria

- 1

Repository records created, updated and last-verified metadata.

Open
- 2

Health flags unstable IDs, missing live controls and likely duplicates.

Open
- 3

Reverification can compare live metadata without overwriting silently.

Open
- 4

Object changes are attributable and dependency-aware.

Open

Implementation coverage

Live highlight can report not found, but no persistent health score, history, similarity detection or change audit exists.

Evidence: ObjectBrowser.tsx current-state stocktake
Last reconciled: 2026-07-29

BL-025 · PB-022 · DATA-01

Partial

Provide a routeable Test Data Library

PRIORITY P1	PLANNED TRANCHE T2 · 2.0 Beta	ACTUAL RELEASE Not yet delivered	WORKSPACE Test Data
ACCEPTANCE 1/4			

Description

Create, find and safely maintain flat datasets with schema and usage information.

User story

As a test-data author, I want a searchable data library so that reusable datasets are governed and understandable.

Acceptance criteria

1	Dataset list supports search, process area, format and stable routes.	Partial
2	Columns have names, types, examples and sensitivity.	Open
3	Rename/removal shows affected Tests and Processes.	Open
4	Scalar, list and table cells survive RFC4180 round-trip.	Implemented

Implementation coverage

Flat CSV create/edit, row operations, rich RFC4180 cells, tags, dirty guards and stable dataset routes exist. Library search/format filters, typed sensitive columns and dependency-safe rename/removal remain.

Evidence: DataEditor.tsx · ListCell.tsx · regression/ui/data-tab.test.js
Last reconciled: 2026-07-29

BL-026 · EXC-009 · EXC-010 · EXC2-002 · SC-05 · SC-06 · SC-17 · SC-18

Implemented

Author nested and relational transaction data

PRIORITY P0	PLANNED TRANCHE T2 · 2.0 Beta	ACTUAL RELEASE Not yet delivered	WORKSPACE Test Data
ACCEPTANCE 4/4			

Description

Design one-header/many-item and many-header/many-item datasets without flattening transaction structure.

User story

As a test-data author, I want nested and relational models so that SAP document loops reflect real business data.

Acceptance criteria

1	UI authors nested JSON transaction arrays.	Implemented
2	UI relates header and child CSVs by declared keys and collection path.	Implemented
3	Duplicates, missing keys, orphans and collection collisions are actionable.	Implemented
4	Preview shows transaction and child counts before save.	Implemented

Implementation coverage

Test Data now authors nested JSON transaction arrays, persists header/child CSV relationship definitions, validates duplicate/missing/orphan/collision conditions through the authoritative loader, and shows transaction plus child counts before save. Joined samples remain nested and owned by their header transaction.

Evidence: packages/core/src/domain/dataSet.ts · packages/studio-server/src/server.ts data preview/relation APIs · packages/studio-web/src/components/DataEditor.tsx · regression/execution-data.test.js · regression/api/data.test.js · regression/ui/data-tab.test.js
Last reconciled: 2026-07-29

BL-027 · EXC-008 · EXC-011 · EXC-019 · EXC2-002 · SC-07 · SC-08 · SC-09 · SC-10 · SC-19 · SC-20

Implemented

Map, preview and snapshot effective execution data

PRIORITY
P0

PLANNED TRANCHE
T2 · 2.0 Beta

ACTUAL RELEASE
Not yet delivered

WORKSPACE
Execution Center, Test Data

ACCEPTANCE
4/4

Description

Bind reusable data to process, stage or Pack member scope and preview exact iterations.

User story

As an operator, I want to inspect effective mappings so that the execution consumes the intended values.

Acceptance criteria

- 1

Mapping supports process data, stage override, Pack-member data, literals, system context and prior outputs.

Implemented
- 2

Filters and maximum limits recalculate the execution matrix.

Implemented
- 3

Preview and Start use the same immutable snapshot.

Implemented
- 4

Invalid mappings block Start and link to the owning workspace.

Implemented

Implementation coverage

The versioned model supports process, stage, Pack-member, literal, system-context and prior-output mappings. Execution Center now authors typed nested-property filters and record limits, recalculates the authoritative matrix, displays every exact selected record plus its content hash and resolved input source, and seals that plan/data into the same immutable snapshot consumed by Start. Filter changes invalidate the preflight token; invalid mappings remain blocking with exact correction routes.

Evidence: packages/core/src/domain/executionPlan.ts applyDataSelection · packages/engine/src/executionOrchestrator.ts · packages/studio-server/src/executionPreflight.ts · packages/studio-web/src/components/RunPanel.tsx · regression/execution-plan-contract.test.js · regression/execution-preflight.test.js · regression/execution-orchestrator.test.js · regression/ui/run-tab.test.js

Last reconciled: 2026-07-29

BL-028 · PB-023 · COL-01 · EXC-001 · SC-04 · SC-11 · SC-12 · SC-13 · SC-14 · SC-15 · SC-16

Implemented

Separate Business Processes and Regression Packs

PRIORITY P1	PLANNED TRANCHE T3 · 2.0 RC	ACTUAL RELEASE 2.1.0	WORKSPACE Processes & Packs
ACCEPTANCE 4/4			

Description

Replace ambiguous Group/Suite concepts with two explicit reusable artifact types.

User story

As a test manager, I want dependency and isolation semantics in business language so that I choose the correct composition.

Acceptance criteria

1	Business Process means ordered dependent Tests with explicit hand-offs.	Implemented
2	Regression Pack means independent Tests or Processes with isolated contexts.	Implemented
3	Legacy Group, Chain, Suite and Batch inputs translate without behavior loss.	Implemented
4	The user-facing workspace is named Processes & Packs after approval.	Implemented

Implementation coverage

The user-facing Processes & Packs workspace now separates ordered Business Processes from persisted Regression Packs. Packs accept independent Test or Business Process members with isolated member data, application, session and failure-policy bindings. Existing Group, Chain, Suite and Batch translation remains backward-compatible.

Evidence: ProcessPacksWorkspace.tsx · PackEditor.tsx · GroupEditor.tsx · server.ts /api/packs · legacyExecutionPlans.ts · packs.test.js · packs-tab.test.js

Last reconciled: 2026-07-30

BL-029 · EXC-018 · EXC2-003

Implemented

Build Processes and Packs visually

PRIORITY
P1

PLANNED TRANCHE
T3 · 2.0 RC

ACTUAL RELEASE
2.1.0

WORKSPACE
Processes & Packs,
Compose

ACCEPTANCE
4/4

Description

Compose sequence, independence, contracts, hand-offs, policies and member bindings without editing plan JSON.

User story

As a process author, I want a visual canvas so that execution topology and data movement are obvious.

Acceptance criteria

- 1

Process canvas shows stages and typed output-to-input hand-offs.

Implemented
- 2

Pack canvas shows independent member lanes and member-specific bindings.

Implemented
- 3

Validation identifies unresolved inputs, cycles, duplicate outputs and incompatible policies.

Implemented
- 4

Keyboard composition, version status and publishing are supported.

Implemented

Implementation coverage

The visual Business Process canvas now persists ordered stages, typed Test contracts and explicit process-data, literal, system-context or prior-stage output bindings. Validation rejects unresolved required inputs, forward or cyclic references, unknown and duplicate outputs, type mismatches and incompatible Pack session policies. Keyboard-native Process and Pack composition, versioned draft/published lifecycles, stable routes and direct mixed Saved Pack execution are covered by isolated API and browser journeys.

Evidence: GroupEditor.tsx visual topology and typed hand-offs · PackEditor.tsx · ProcessPacksWorkspace.tsx · server.ts /api/groups and /api/packs validation · executionPreflight.ts saved Pack planning · packages/cli/src/commands/batch.ts · groups.test.js · packs.test.js · runs.test.js · groups-tab.test.js · packs-tab.test.js · run-tab.test.js

Last reconciled: 2026-07-30

BL-030 · PB-024 · EXE-01 · EXC-017 · EXC-020 · EXC-021

Implemented

Complete authoritative preflight and execution review

PRIORITY P0	PLANNED TRANCHE T2 · 2.0 Beta	ACTUAL RELEASE Not yet delivered	WORKSPACE Execution Center
ACCEPTANCE 4/4			

Description

Validate target, artifacts, data, policies and calculated work through one server-owned model.

User story

As an operator, I want authoritative preflight so that configuration errors are corrected before SAP opens.

Acceptance criteria

1	Server validates target, credentials, versions, contracts, objects, mappings and policies.	Implemented
2	Review shows members, iterations, stages, steps and known child work.	Implemented
3	Blocking findings link to the owning artifact; warnings require acknowledgement.	Implemented
4	Start token and plan hash prevent stale or duplicate starts.	Implemented

Implementation coverage

Authoritative server preflight validates target freshness, credential availability, plan/contract versions, repository objects, data mappings and execution policies. Review presents the full calculated matrix and child count; correction routes and warning acknowledgement are enforced; token, plan hash, target hash and duplicate-claim handling protect Start.

Evidence: executionPreflight.ts · RunPanel.tsx · regression/execution-preflight.test.js · regression/ui/run-tab.test.js

Last reconciled: 2026-07-29

BL-031 · PB-025 · RUN-01 · EXC-002 · EXC-003 · EXC-013 · EXC-022 · EXC-027 · EXC2-004 · SC-22

Partial

Complete hierarchical execution monitoring

PRIORITY P1	PLANNED TRANCHE T3 · 2.0 RC	ACTUAL RELEASE Not yet delivered	WORKSPACE Execution Center
ACCEPTANCE 2/4			

Description

Monitor Pack, Process, Test, Iteration, Step and Child Item from persisted events.

User story

As an operator, I want the complete hierarchy so that progress and failures never appear flattened or stalled.

Acceptance criteria

1	All six hierarchy levels have stable identity, status, totals and duration.	Partial
2	Known child work contributes measurable progress.	Implemented
3	Refresh/reconnect rebuilds the same state from persisted events.	Implemented
4	Monitor, history, metrics and evidence counts reconcile.	Open

Implementation coverage

Stable run routes, persisted run/progress records, reconnect recovery and measurable child-work progress exist. A complete six-level persisted event hierarchy with stable child identities and cross-surface count reconciliation remains.

Evidence: RunPanel.tsx · runs.ts · regression/execution-child-progress.test.js
Last reconciled: 2026-07-29

BL-032 · PB-026 · FAIL-01 · EXC-023

Partial

Provide focused failure diagnosis

PRIORITY P1	PLANNED TRANCHE T4 · 2.0 GA	ACTUAL RELEASE Not yet delivered	WORKSPACE Execution Center, Audit & Evidence
ACCEPTANCE 3/4			

Description

Explain the first actionable failure with safe context and links to correction.

User story

As an investigator, I want focused diagnosis so that I can correct the cause without reading raw logs.

Acceptance criteria

1	Failure location identifies Process, Test, iteration, step and child item.	Implemented
2	Expected/actual context is redacted according to policy.	Implemented
3	Links open the exact Test, object, dataset, mapping or setting.	Open
4	Raw diagnostics remain secondary and safe.	Implemented

Implementation coverage

Focused diagnosis identifies member/process, iteration, stage, step and child work; secret filtering/redaction is enforced and raw logs remain secondary. Artifact-specific correction links from a terminal failure remain.

Evidence: RunPanel.tsx result rendering · EXC-023
Last reconciled: 2026-07-29

BL-033 · PB-027 · RERUN-01 · EXC-016 · EXC2-005 · SC-21

Implemented

Rerun safely with lineage and explicit differences

PRIORITY P0	PLANNED TRANCHE T4 · 2.0 GA	ACTUAL RELEASE Not yet delivered	WORKSPACE Execution Center, Audit & Evidence
ACCEPTANCE 4/4			

Description

Review failed/full rerun scope and compare it with the immutable source execution.

User story

As an operator, I want traceable reruns so that recovery does not duplicate business side effects.

Acceptance criteria

1	Rerun review compares plan, data, policies, target and scope.	Implemented
2	Failed-only rerun is offered only where runtime semantics permit.	Implemented
3	New run records parent, reason, scope and changed inputs.	Implemented
4	Cancellation, retry and resume do not duplicate completed side effects.	Implemented

Implementation coverage

Rerun is now a two-step review-and-confirm workflow that compares immutable plan, selected data, session/failure policies, current verified target and requested scope. The review hash prevents post-review drift; the new run persists parent, reason, scope, reviewed hash and every changed input. Failed-only scope is disabled when no eligible work exists, excludes passed records even inside a failed Pack member, and blocks any started transactional iteration whose SAP state must be retained. Full transactional replay is likewise blocked when it could duplicate documents. Cancellation remains at the active-transaction boundary and request keys make retry idempotent.

Evidence: packages/core/src/domain/executionPlan.ts transaction policy snapshot · packages/studio-server/src/runs.ts reviewRerun and createRerunSnapshot · packages/studio-server/src/server.ts rerun-review endpoint · packages/studio-web/src/components/RunPanel.tsx rerun difference review · regression/execution-recovery.test.js · regression/api/runs.test.js · regression/ui/run-tab.test.js

Last reconciled: 2026-07-29

BL-034 · PB-028 · EXC-024 · EXC2-001 · SC-26

Implemented

Secure canonical evidence and provenance

PRIORITY P0	PLANNED TRANCHE T1 · 2.0 Alpha	ACTUAL RELEASE Not yet delivered	WORKSPACE Audit & Evidence, Execution Center
ACCEPTANCE 4/4			

Description

Protect one immutable evidence source and expose safe provenance from both run and audit views.

User story

As an auditor, I want secure canonical evidence so that records are trustworthy and not duplicated.

Acceptance criteria

1	Execution Center and Audit resolve the same archived evidence artifact.	Implemented
2	Artifact access requires authenticated owner authorization.	Implemented
3	Evidence shows run, target class, time zone, plan/data versions and redaction state.	Implemented
4	Missing evidence explains its lifecycle reason without leaking existence.	Implemented

Implementation coverage

Execution Center and Audit reuse one owner-protected archive artifact. New PDFs record run/execution identity, target class and verification time, time zone, plan/snapshot/data versions and enforced redaction state; the Audit library presents retention/redaction policy and a safe lifecycle explanation when no PDF exists.

Evidence: evidencePdf.ts provenance metadata · server.ts protected artifact mounts · /api/evidence-governance · DocumentsPanel.tsx · evidence-reference.test.js · t1-security-context.test.js
Last reconciled: 2026-07-29

BL-035 · PB-029 · HIST-01 · EXC-024

Partial

Complete searchable Audit and Evidence history

PRIORITY P1	PLANNED TRANCHE T4 · 2.0 GA	ACTUAL RELEASE Not yet delivered	WORKSPACE Audit & Evidence
ACCEPTANCE 1/4			

Description

Find immutable runs, inspect stable details and traverse source/rerun lineage.

User story

As a test manager or auditor, I want meaningful history search so that I can answer delivery and compliance questions.

Acceptance criteria

1	Server supports date, outcome, artifact, App ID, environment, run ID and executor filters.	Partial
2	Results paginate and sort by start, duration and status.	Partial
3	Every result opens a stable detail route with parent/child lineage.	Partial
4	Evidence types and source artifacts are linked from the run record.	Implemented

Implementation coverage

Flat newest-first history, client filters, stable audit detail routes and canonical PDF/source fields exist. Server-side multi-field filtering, pagination/sorting and complete parent/child lineage traversal remain.

Evidence: DocumentsPanel.tsx · regression/ui/audit-library.test.js

Last reconciled: 2026-07-29

BL-036 · PB-030 · SET-01 · EXC-025 · EXC2-001

Implemented

Secure identity, target and integration administration

PRIORITY P0	PLANNED TRANCHE T1 · 2.0 Alpha	ACTUAL RELEASE Not yet delivered	WORKSPACE Settings, Shell, Execution Center, Control Object Repository
ACCEPTANCE 4/4			

Description

Protect the sole-owner workspace and server-side SAP integration without exposing credentials.

User story

As the workspace owner, I want secure integration settings so that execution and scanning use known targets safely.

Acceptance criteria

1	Authenticated owner identity protects mutations, runs and artifacts.	Implemented
2	Credentials remain server-side and are never returned, logged or committed.	Implemented
3	SAP target has URL, profile reference, safety class and verification status.	Implemented
4	Existing workspace data remains visible when the owner links a Google account.	Implemented

Implementation coverage

Single-owner authentication protects APIs and artifacts; Google linking retains the existing workspace. Credentials stay in the server-side vault/encrypted fallback and are redacted from logs. Settings now persists explicit target classification and performs a non-transactional login-only verification; executions and reruns capture the initiating owner and verified target snapshot.

Evidence: auth.ts · workspaceGovernance.ts · sapVerification.ts · SettingsPanel.tsx · runs.ts · CLI --executed-by and target provenance · sap-settings.test.js · t1-security-context.test.js
Last reconciled: 2026-07-29

BL-037 · PB-033 · FUT-02 · New cross-workspace requirement

Not started

Add global artifact search and dependency impact

PRIORITY P2	PLANNED TRANCHE T5 · Post-2.0	ACTUAL RELEASE Not yet delivered	WORKSPACE Automation Overview, Compose, Test Data, Processes & Packs, Control Object Repository, Audit & Evidence
ACCEPTANCE 0/4			

Description

Search Tests, objects, datasets, Processes, Packs and runs and understand their relationships.

User story

As a maintainer, I want global search and dependency information so that changes are safe and duplicates are avoided.

Acceptance criteria

1	Search returns typed results with domain, application and lifecycle status.	Open
2	Detail views show verified incoming and outgoing usage.	Open
3	Rename, delete and archive use dependency information.	Open
4	Search respects owner and future role authorization.	Open

Implementation coverage

Each workspace has isolated lists or filters; no cross-artifact index or dependency graph exists.

Evidence: Current workspace implementation stocktake

Last reconciled: 2026-07-29

BL-038 · FND-08 · EXC-004 · EXC-005 · EXC-012 · EXC-014 · EXC-015 · EXC-028 · EXC2-006 · SC-23 · SC-24 · SC-25

Implemented

Protect compatibility, migration and execution correctness

PRIORITY P0	PLANNED TRANCHE T4 · 2.0 GA	ACTUAL RELEASE Not yet delivered	WORKSPACE Quality Engineering, Execution Center
ACCEPTANCE 4/4			

Description

Retain legacy behavior through one versioned orchestration model and comprehensive regression coverage.

User story

As a product owner, I want migration and correctness tests so that redesign does not change intended execution behavior.

Acceptance criteria

1	Legacy Chain, Suite and Batch translate to versioned plans.	Implemented
2	Every selected Batch row executes and transaction outputs remain isolated.	Implemented
3	Policies, Pack scheduling, cancellation and evidence counts have regression coverage.	Implemented
4	Build, typecheck, isolated UI, secret scan and approved live cases pass before GA.	Implemented

Implementation coverage

Versioned translation, unified orchestration, every-row Batch behavior, isolation, policies, cancellation and evidence counts have regression coverage. The repository-wide migration audit translates every persisted Test and Group without source mutation, validates the combined Batch Pack, and preserves Suite order and independent identity. Build/typecheck, 57 core tests, 28 isolated API tests, 32 isolated UI tests and the high-confidence secret scan are green. Read-only live modes, controlled headed P2P/O2C and the authorised negative remediation are recorded. Run 299669f8-75b6-471d-9c56-ed0375011fc5 passed with immutable retained-state evidence, completing the approved live cases and criterion 4.

Evidence: executionPlan.ts · legacyExecutionPlans.ts · regression/legacy-migration.test.js · regression/secret-scan.mjs · package.json
test:release:isolated · regression/results/quality-history.json · audit-evidence/299669f8-75b6-471d-9c56-ed0375011fc5/evidence.pdf · QA/4HANA Test Operations catalogue
Last reconciled: 2026-07-30

BL-039 · PB-032 · FUT-01

Deferred

Add multi-user roles and capabilities

PRIORITY P3	PLANNED TRANCHE T5 · Post-2.0	ACTUAL RELEASE Not yet delivered	WORKSPACE Platform, All workspaces
ACCEPTANCE Deferred			

Description

Introduce server-enforced permissions only when multi-user deployment is approved.

User story

As a product owner, I want role-based capabilities so that users see and perform authorised actions.

Acceptance criteria

1	Implementation starts after a confirmed multi-user requirement.	Deferred
2	Server protects every mutation, execution, artifact and administration action.	Deferred
3	Role mapping covers manager, consultant, engineer, tester, analyst and auditor.	Deferred

Implementation coverage

The product intentionally uses one preserved owner identity for 2.0.

Evidence: Single-owner product decision

Last reconciled: 2026-07-29

BL-040 · FUT-04

Deferred

Add scheduling, notification and controlled parallelism

PRIORITY P3	PLANNED TRANCHE T5 · Post-2.0	ACTUAL RELEASE Not yet delivered	WORKSPACE Execution Center, Automation Overview
ACCEPTANCE Deferred			

Description

Schedule approved work, notify outcomes and parallelise only where isolation is proven.

User story

As a test manager, I want scheduled and parallel runs so that regression throughput scales safely.

Acceptance criteria

1	Schedules capture immutable scope, target and owner.	Deferred
2	Notifications link to stable run records without exposing secrets.	Deferred
3	Parallel work respects target capacity, data isolation and cancellation.	Deferred
4	Concurrency limits and queue health are observable.	Deferred

Implementation coverage

Deliberately outside the 2.0 release; Pack member scheduling semantics exist only for current sequential orchestration.

Evidence: Approved 2.0 scope decision

Last reconciled: 2026-07-29

BL-041 · T1.3.1 · Execution validation closure

Implemented

Close authoritative Chain, Suite and Batch validation

PRIORITY P0	PLANNED TRANCHE T1 · 2.0 Alpha	ACTUAL RELEASE Not yet delivered	WORKSPACE Quality Engineering, Execution Center
ACCEPTANCE 4/4			

Description

Replace assumed or skipped execution coverage with recorded isolated and non-production SAP results while separating read-only validation from business-document transactions.

User story

As the product owner, I want trustworthy mode-level test evidence so that T1.4 starts only after the execution heart of the product is proven.

Acceptance criteria

1	The Test Operations catalogue is generated from recorded test-run output and never fabricates status, time or failure history.	Implemented
2	Isolated API and Studio UI journeys complete Chain, Suite and multi-group Batch with synthetic execution and no SAP side effects.	Implemented
3	Read-only API and Studio UI journeys complete Chain, Suite and two-group Batch against a live-verified non-production SAP target.	Implemented
4	Transactional P2P and negative cases have approved test-data ownership, unique run references, fail-stop behavior, retained-state evidence and a recorded pass. Any remediation is a separate explicitly authorized activity.	Implemented

Implementation coverage

The 125-test Test Operations catalogue is generated from repository inventory plus recorded Node output; executable reconciliation proves that status, timestamp, duration and retained failure history are not fabricated. Core (57), isolated API (28), isolated UI (32), live read-only API (3 modes) and live read-only Studio UI (3 modes) are green. Controlled headed P2P and O2C runs passed with owner-linked references, fail-stop policy and canonical evidence. The first controlled negative attempt, eb060bc6-673b-4e65-8919-6ccf66b6a645, remains immutably failed and retained. After explicit owner authorisation, remediation run 299669f8-75b6-471d-9c56-ed0375011fc5 passed all six steps under reference Q4HP2P2607309366: the title remained New Purchase Order, SAP displayed "Document contains no items," no document number or downstream step occurred, and state remained retain-for-review. The recorded pass is ingested into Test Operations. All four criteria are complete.

Evidence: regression/reporters/result-capture.js · regression/record-quality-run.mjs · regression/quality-catalog.test.js · regression/results/quality-history.json · regression/results/raw/live-negative-remediation-2026-07-30.ndjson · regression/transactional-negative.test.js · packages/engine/src/modules/assertDocumentCreationBlocked.ts · testcases/create-po-missing-item-negative.json · regression/api/runs.test.js · regression/ui/run-tab.test.js · audit-evidence/d55cd0f0-07ed-433f-8e40-1b61cd102827/evidence.pdf · audit-evidence/4b4f15d7-32df-4348-be70-

[23fe5243ff7a/evidence.pdf](#) · [audit-evidence/eb060bc6-673b-4e65-8919-6ccf66b6a645/evidence.pdf](#) · [audit-evidence/299669f8-75b6-471d-9c56-ed0375011fc5/evidence.pdf](#) · QA/4HANA Test Operations catalogue

Last reconciled: 2026-07-30

RECORDED FAILURE LEDGER

4 Defects and issues fixed

The product test ledger preserves failures even after remediation. All 16 recorded failures are remediated and linked to a subsequent verifying run.

ID / status	Area / mode	Observed failure	Resolution evidence
FL-0016 Remediated	Process Suites Live UI · Non-production SAP	live: Run tab read-only Batch runs two independent SAP smoke groups expected both groups to pass, got: "✓ All 2 process iterations passed exit code 0 2 of 2 process iterations completed 100%"	Reconciled the UI assertion with hierarchical Batch completion wording and reverified both live smoke groups. Verified: T1.3.1-LIVE-UI-R2
FL-0015 Remediated	Test Data Live UI · Non-production SAP	Run tab: a preflight data finding opens the exact dataset route locator.click: Timeout 30000ms exceeded. Call log: - waiting for locator('li:has-text("route-mapped.json") button:has-text("+ Add")')	Aligned the correction journey with the route-aware dataset selection flow and reverified the exact dataset route. Verified: T1.3.1-ISOLATED-UI-R4
FL-0014 Remediated	Compose API · Synthetic non-production	GET /api/testcases/:file returns the parsed test case Expected values to be strictly equal: 404 != 200	Corrected isolated Test fixture provisioning so the detail endpoint returns the persisted Test. Verified: T1.3.1-ISOLATED-API-R2
FL-0013 Remediated	Compose API · Synthetic non-production	GET /api/testcases lists known fixtures expected create-po.json in the list	Corrected isolated Test fixture discovery so known Tests appear in the list. Verified: T1.3.1-ISOLATED-API-R2
FL-0012 Remediated	Process Suites API · Synthetic non-production	GET /api/groups/:file returns the parsed group Expected values to be strictly equal: + actual - expected + 'Synthetic Process' - 'Create PO - GR - Invoice'	Corrected isolated Business Process fixture identity and response expectations. Verified: T1.3.1-ISOLATED-API-R2
FL-0011 Remediated	Process Suites API · Synthetic non-production	GET /api/groups lists known fixtures The expression evaluated to a falsy value: assert.ok(body.includes('o2c-e2e.json'))	Corrected isolated Business Process fixture discovery for the O2C process. Verified: T1.3.1-ISOLATED-API-R2
FL-0010 Remediated	Test Data API · Synthetic non-production	GET /api/data/:file parses an existing CSV The expression evaluated to a falsy value: assert.ok(body.headers.includes('supplier'))	Corrected isolated CSV fixture provisioning and header expectations. Verified: T1.3.1-ISOLATED-API-R2
FL-0009 Remediated	Test Data API · Synthetic non-production	GET /api/data lists known fixtures The expression evaluated to a falsy value: assert.ok(body.includes('p2p-e2e.csv'))	Corrected isolated data fixture discovery for P2P data. Verified: T1.3.1-ISOLATED-API-R2
FL-0008 Remediated	Process Suites Headless UI · Synthetic non-production	Run tab: Pack Processes completes a synthetic multi-group Batch expected two Batch groups to pass, got: "✓ All 2 process iterations passed exit code 2 of 2 process iterations completed 100%"	Reconciled Pack · Processes completion output with the two-member hierarchical result. Verified: T1.3.1-ISOLATED-UI-R3

ID / status	Area / mode	Observed failure	Resolution evidence
FL-0007 Remediated	Execution Center Headless UI · Synthetic non-production	Run tab: Pack Tests completes a synthetic independent Suite Run did not complete within 30000ms	Stabilized synthetic Pack · Tests execution completion and polling within the supported run lifecycle. Verified: T1.3.1-ISOLATED-UI-R3
FL-0006 Remediated	Process Suites Headless UI · Synthetic non-production	Run tab: Business Process completes a synthetic multi-stage Chain Run did not complete within 30000ms	Stabilized synthetic multi-stage Business Process completion and polling. Verified: T1.3.1-ISOLATED-UI-R3
FL-0005 Remediated	Process Suites Headless UI · Synthetic non-production	Run tab: Pack Processes completes a synthetic multi-group Batch [object Object]	Corrected the synthetic Pack · Processes execution path and verified two independent members. Verified: T1.3.1-ISOLATED-UI-R3
FL-0004 Remediated	Execution Center Headless UI · Synthetic non-production	Run tab: Batch mode runs "Cleanup Drafts" to completion (execution opt-in) [object Object]	Corrected the synthetic Batch compatibility execution path and completion state. Verified: T1.3.1-ISOLATED-UI-R2
FL-0003 Remediated	Execution Center Headless UI · Synthetic non-production	Run tab: Pack Tests completes a synthetic independent Suite [object Object]	Corrected the synthetic Pack · Tests execution path and independent member results. Verified: T1.3.1-ISOLATED-UI-R3
FL-0002 Remediated	Process Suites Headless UI · Synthetic non-production	Run tab: Business Process completes a synthetic multi-stage Chain [object Object]	Corrected the synthetic Business Process execution path and ordered stage results. Verified: T1.3.1-ISOLATED-UI-R3
FL-0001 Remediated	Automation Overview Headless UI · Synthetic non-production	Canvas First Overview presents the approved shell and real workspace data [object Object]	Aligned the Overview fixture/API state with the approved Canvas First metrics and shell. Verified: T1.3.1-ISOLATED-UI-R2

Open defect count: zero in the recorded ledger. This statement is limited to the repository-backed failure ledger as of 30 July 2026; newly reported production observations must still be logged and triaged.

REMAINING ACCEPTANCE SCOPE

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Items requiring remediation or future delivery

These are product gaps or intentionally deferred capabilities, not unverified claims of production defects.

ID / status	Item	Criteria still requiring work	Recommended remediation
BL-018 Partial	Complete the task-focused Automation Overview Automation Overview · P1	- Selected tests and runs open their exact detail routes. - Recent failures, stale drafts and blocked readiness are surfaced without duplicating navigation.	Make every recent item route to its exact artifact or run and add failure, stale-draft and readiness attention queues.
BL-019 Partial	Add filterable execution value and trend analytics Automation Overview · P1	- Every metric states date range, filters, numerator, denominator and freshness. - Cost assumptions are saved as owner workspace preferences. - Charts have accessible tabular equivalents and never fabricate trends.	Persist owner cost assumptions, add explicit metric scope/freshness, filtered trends and accessible tables.
BL-022 Partial	Complete the Control Object Repository workbench Control Object Repository · P1	- Search spans domain, App ID, name, label, type and stability. - Object detail shows selector metadata, scope, usage and last verification. - Rename and delete show dependency impact and preserve or guide reference correction.	Add selector stability, usage and verification metadata plus dependency-safe rename and removal.
BL-023 Partial	Complete guided live-screen capture Control Object Repository, Compose · P1	Contextual capture returns the saved object to the originating Compose field without losing work.	Carry the originating Compose field through live capture and return the saved object without losing edits.
BL-024 Not started	Add selector health, history and governance Control Object Repository, Audit & Evidence · P1	- Repository records created, updated and last-verified metadata. - Health flags unstable IDs, missing live controls and likely duplicates. - Reverification can compare live metadata without overwriting silently. - Object changes are attributable and dependency-aware.	Create selector health scoring, history, live comparison, similarity detection and attributable change audit.

ID / status	Item	Criteria still requiring work	Recommended remediation
BL-025 Partial	Provide a routeable Test Data Library Test Data · P1	- Dataset list supports search, process area, format and stable routes. - Columns have names, types, examples and sensitivity. - Rename/removal shows affected Tests and Processes.	Build the dataset library, typed/sensitive columns and dependency impact for rename or removal.
BL-031 Partial	Complete hierarchical execution monitoring Execution Center · P1	- All six hierarchy levels have stable identity, status, totals and duration. - Monitor, history, metrics and evidence counts reconcile.	Persist the complete six-level event hierarchy and reconcile child totals across monitor, history, metrics and evidence.
BL-032 Partial	Provide focused failure diagnosis Execution Center, Audit & Evidence · P1	Links open the exact Test, object, dataset, mapping or setting.	Add exact terminal-failure correction links to the owning Test, object, dataset, mapping or setting.
BL-035 Partial	Complete searchable Audit and Evidence history Audit & Evidence · P1	- Server supports date, outcome, artifact, App ID, environment, run ID and executor filters. - Results paginate and sort by start, duration and status. - Every result opens a stable detail route with parent/child lineage.	Move filters, pagination, sorting and complete parent/child lineage traversal to the server.
BL-037 Not started	Add global artifact search and dependency impact Automation Overview, Compose, Test Data, Processes & Packs, Control Object Repository, Audit & Evidence · P2	- Search returns typed results with domain, application and lifecycle status. - Detail views show verified incoming and outgoing usage. - Rename, delete and archive use dependency information. - Search respects owner and future role authorization.	Build an authorized cross-artifact index and dependency graph before exposing global search.
BL-039 Deferred	Add multi-user roles and capabilities Platform, All workspaces · P3	- Implementation starts after a confirmed multi-user requirement. - Server protects every mutation, execution, artifact and administration action. - Role mapping covers manager, consultant, engineer, tester, analyst and auditor.	Keep deferred until multi-user requirements and role ownership are explicitly approved.

ID / status	Item	Criteria still requiring work	Recommended remediation
BL-040 Deferred	Add scheduling, notification and controlled parallelism Execution Center, Automation Overview · P3	• Schedules capture immutable scope, target and owner. • Notifications link to stable run records without exposing secrets. • Parallel work respects target capacity, data isolation and cancellation. • Concurrency limits and queue health are observable.	Keep deferred until schedules, notifications, target capacity and concurrency governance are designed.

DELIVERY GUIDANCE

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Verification evidence and recommended next sequence

Evidence base used

- Authoritative HTML Product Backlog Tracker.
- Repository-generated 149-test operations catalogue.
- Recorded execution and failure history.
- Core, API, UI, accessibility, secret-scan and live-run documentation.
- Current production build source and anonymized workspace captures.

Recommended next implementation sequence

1. BL-018 and BL-019: complete Overview deep links, alerts and governed analytics.
2. BL-022 through BL-025: finish object and data governance foundations.
3. BL-031, BL-032 and BL-035: close execution hierarchy, diagnosis and audit traversal.
4. BL-037: add global search only after dependency metadata is reliable.
5. Retain BL-039 and BL-040 as deferred until multi-user and scheduling scope is approved.

Change-control recommendation. Update the HTML tracker and regenerate this report after each accepted backlog tranche. Record every failed automated case in the durable ledger even when it is subsequently remediated.